

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		Use of value taken from graph within < 0.1 at 2.0 m [expect 4.4 m/s] (1) Substitution: wavelength = $\frac{4.4}{0.2}$ (1) =22.0 m (1) Use of $v = 3.13\sqrt{d}$ to give 4.43 m/s → lose first mark	1	1 1		3	3	
		(ii)		If frequency is constant (1) as speed increases wavelength increases (1) Alternative [by calculation] Using 0.2 Hz (1) and calculating wavelength at a different stated depth (1) To award 2 marks 'suggestion not true' [or equiv] must be made			2	2		
				Question 1 total	4	7	4	15	11	0